

Experiment No. 2

Program:

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
import string

# Run only once
nltk.download("punkt")
nltk.download("stopwords")

text = "Natural Language Processing is an exciting field of Artificial Intelligence."

print("Original Text:")
print(text)

# Tokenization
tokens = word_tokenize(text)
print("\nTokens:")
print(tokens)

# Filtration
filtered = [
    word for word in tokens
    if word not in string.punctuation
    and word.lower() not in stopwords.words("english")
]

print("\nFiltered Tokens:")
print(filtered)

# Script Validation
print("\nScript Validation:")
print(["English (Latin Script)" if text.isascii() else "Non-English Script"])
```

Output:

```
Original Text:
Natural Language Processing is an exciting field of Artificial Intelligence.

Tokens:
['Natural', 'Language', 'Processing', 'is', 'an', 'exciting', 'field', 'of', 'Artificial', 'Intelligence', '.']

Filtered Tokens:
['Natural', 'Language', 'Processing', 'exciting', 'field', 'Artificial', 'Intelligence']

Script Validation:
English (Latin Script)
```

Experiment No. 3

Program:

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer
import string

# Run only once
nltk.download("punkt")
nltk.download("stopwords")
nltk.download("wordnet")

text = "The students are studying Natural Language Processing and playing with computers."

print("Original Text:")
print(text)

tokens = [
    word for word in word_tokenize(text)
    if word not in string.punctuation
]

filtered = [
    word for word in tokens
    if word.lower() not in stopwords.words("english")
]

print("\nAfter Stop Word Removal:")
print(filtered)

stemmer = PorterStemmer()
print("\nAfter Stemming:")
print([stemmer.stem(word) for word in filtered])

lemmatizer = WordNetLemmatizer()
print("\nAfter Lemmatization:")
print([lemmatizer.lemmatize(word) for word in filtered])
```

Output:

Original Text:

The students are studying Natural Language Processing and playing with computers.

After Stop Word Removal:

['students', 'studying', 'Natural', 'Language', 'Processing', 'playing', 'computers']

After Stemming:

['student', 'studi', 'natur', 'languag', 'process', 'play', 'comput']

After Lemmatization:

['student', 'studying', 'Natural', 'Language', 'Processing', 'playing', 'computer']